Lab 4

Introduction to Databases

Constraints

- **NOT NULL :** Must have a value
- **PRIMARY KEY :** Primary Key (automatically NOT NULL + UNIQUE)
- **FOREIGN KEY :** used to combine 2 table, one table used as a reference to the other table
- **UNIQUE :** Cannot repeat

Constraint	Column-Level Syntax	Table-Level Syntax
NOT NULL	column datatype CONSTRAINT constraint_name NOT NULL	Not Allowed
PRIMARY KEY	column datatype CONSTRAINT constraint_name PRIMARY KEY	CONSTRAINT constraint_name PRIMARY KEY (column)
UNIQUE	column datatype CONSTRAINT constraint_name UNIQUE	CONSTRAINT constraint_name UNIQUE (column)
FOREIGN KEY	column datatype CONSTRAINT constraint_name REFERENCES parent_table(parent_column)	CONSTRAINT constraint_name FOREIGN KEY (column) REFERENCES parent_table(parent_column)

SQL command exercises

1. Create a Table: Create a table STUDENTS with the following columns:

- id → NUMBER (Primary Key)
- name → VARCHAR2(50) Not Null
- email→ VARCHAR2(100) UNIQUE
- City → VARCHAR2(50) Not Null
- age → NUMBER

2. Check constraint (Before create the table): the minimum age is 18 and the maximum is 50.

CONSTRAINT constraint_name Check (column_name condition)

3. ALTER TABLE (After create the table): Add UNIQUE constraint to mobile

: check constraint of city to be (Cairo, Alex, Aswan)

4. Drop/disable Constraints: Drop / Disable the age constraints

drop CONSTRAINT constraint_name

disable CONSTRAINT constraint_name

5. Insert Data into a Table: Insert the following students into the table:

ID	Name	Email	City	Age	Mobile
1	Ali	ali@gmail.com	Cairo	60	2667
1	Ahmed	ahmed@gmail.com	Aswan	21	1239
2	Sara	sara@gmail.com	Mansoura	22	2265
3	NULL		Alex	30	2364

6. Update Data into a Table: Update age of student with ID = 1 to 23
: Update Email of student with ID=1 to 'sara@gmail.com'

7. Delete Data from a Table: Delete student with ID = 3

8. Drop: Delete the table completely

Oracle Retrieving Data

- **The Select Statement**
- **The Where clause**
- **Combine conditions**
- **Using Like, IN, Between and Not**

1. Create a Table: Create a table Students with the following columns:

- id → NUMBER (Primary Key)
- name → VARCHAR2(50) Not Null
- email→ VARCHAR2(100) UNIQUE
- City → VARCHAR2(50)
- age → NUMBER
- Grade → NUMBER

2. Insert 5 Samples Data

ID	Name	Email	Age	City	Score
1	Ali	ali@gmail.com	20	Cairo	85
2	Sara	sara@yahoo.com	22	Alex	90
3	Omar	omar@gmail.com	19	Cairo	70
4	Mona	mona@hotmail.com	21	Giza	95
5	Hany	hany@gmail.com	23	Alex	60

3. Select all data in the table

4. Select the Students from Cairo

5. Students from Cairo AND grade > 80

6. Students from Cairo OR Alex

7. Names starting with 'A'

8. Emails ending with gmail.com

9. Names with 4 letters

10. Age between 20 and 22

11. Students from Cairo or Giza

12. Students NOT from Alex

13. Students from Cairo AND age between 18 and 21

14. return only the NAME and EMAIL of students older than 18.

Home work

Q1. Create table DEPARTMENTS with:

Column	Datatype	Constraint
DEPT_ID	NUMBER	Primary Key
DEPT_NAME	VARCHAR2(50)	Not Null

Q2. Create table STUDENTS with:

Column	Datatype	Constraint
ID	NUMBER	Primary Key
NAME	VARCHAR2(50)	Not Null
EMAIL	VARCHAR2(100)	Unique
AGE	NUMBER	Between 18 and 30
DEPT_ID	NUMBER	Foreign Key references DEPARTMENTS

Q3. Insert:

- 2 departments
- 3 students (valid data)

Q4.

- Update a student’s AGE to 25
- Try to update AGE to 40 → What happens? Why?
- Update student EMAIL

Q5.

- Add column PHONE VARCHAR2(15)
- Add NOT NULL constraint to PHONE
- Modify AGE datatype to NUMBER(3)
- Drop UNIQUE constraint from EMAIL
- Retrieve the email if the age less then 40

Q6.

- Drop CHECK constraint on AGE
- Drop STUDENTS table
- Drop DEPARTMENTS table